

Workstation

Quadro and FirePro are, respectively, the brands that NVIDIA and AMD have in this space. The most prominent difference between gaming graphics cards and workstation cards is that workstation cards focus on OpenGL libraries while gaming cards have a lot more on their plate with DirectX, Mantle, Vulkan etc. Workstation machines also work on textures which are of much higher resolution than what you see in video games. This means they have a lot more RAM to handle humongous calculations so that the data need not be sent to the system RAM. Just to put things into perspective, dual-display setups are commonplace with each monitor rendering frames at 8K resolution. Whatever you enjoy on your gaming card was actually made on a workstation card and given their capabilities, these cards cost a bomb. Another key difference is that, these cards can crunch a lot more double-precision Floating Point (FP) numbers while their gaming counterparts focus on single-precision FP numbers.

What do the numbers mean?

Fortunately, testing graphics cards is simple when you are looking at benchmark scores. Understanding the core architecture can be an uphill task since it does require a fair bit of understanding of the underlying technologies that are used in microprocessors. So what kind of numbers are you going to be looking at?

FPS

Frame-per-second is the prime metric that you'll be looking at when it comes to games. There are a few thresholds that you should keep in mind. 24FPS is what your eyes can perceive without any problems but 30FPS is preferred over 24 since minor stuttering that occur during intensive scenes go unnoticed if your average FPS scores are at 30. 60FPS is the holy grail that most gamers aspire towards since in all likelihood if your average score is 60FPS then the most intensive scenes will probably be above 24FPS. Anything that goes lower is going to seem stressful with lower numbers being really painful to watch. These days we have monitors that synchronise with the graphics cards to lessen the effects of vastly varying frame rates.

HOW TO TEST MORE

We use a good number of games as well as synthetic benchmark suites to generate scores and some of the software used are quite expensive for an individual to buy for a one-off test. So before we get into the different benchmarks that are available, we need to establish a few ground rules. You can call them common benchmark practices. Firstly, you need to ensure that no new software are installed between benchmarking two different graphics cards. Each new software added might have a background process which will eat up a few CPU cycles bringing about a small difference in scores. It could be negligible in most cases but we don't take any chances. The same goes for drivers, so ensure the same set is used for all cards you test but there will be instances where upgrading the driver is necessary since certain drivers bring noticeable improvements in scores. To compare your scores with other cards, you need to pick a popular website like www.digit.in (*wink wink*) and compare the score obtained in a particular game and run the same at your end. There will be a difference if your PC configuration is different than what the site uses and you can use this score to add or subtract a percentage to normalise scores. There can be a variation in scores if you are using a different version of the game or benchmark, so normalisation is something you will have to do.

Matching temperature scores is very difficult since the variable parameters are too darn many. We only look towards maintaining the ambient temperature and you should do the same too. A variation of a few degrees between the site's reading and yours is fine but anything in double digits is reason enough for investigation.

There is an entire bunch of parameters that go into scoring the build of the card. Some of which include the different materials that go into the chassis construction, the cooling mechanism, the choice of material, the placement of components on the circuit board, the accessibility of switches if present, the slight droop of the card when slotted sideways, the efficacy of unique features etc.

The benchmarks are chosen with an even distribution of proprietary APIs so that one competitor doesn't end up getting better scores than the other. Each benchmark is run multiple times over to get an average score. This eliminates any instances where the score might have dropped owing to background processes of the operating system that we have no control over.

And lastly, each routine of benchmarks are run at different resolutions under two different states. Once when the graphics card is set to factory clock speeds and once when it has been overclocked. You don't need to run the entire gamut of our benchmarks, only the ones that you are bothered with. Otherwise, you are looking at a few weeks worth of benchmarking.

Watts

If you are worried about long term cost overheads of investing in a powerful card then this metric is what you should be concerned with. There are plenty of bundled software that record wattage levels over time. So you can estimate as to how much more are you going to have to pay in terms of electricity consumption.

Temperature

Graphics cards get hot, really hot, so temperature is still a factor, but one that will soon be of less relevance owing to improvements in chipset architecture. The problem with heat is that it lessens the lifespan of your graphics card as well as the components around it if not taken care of.

Synthetic scores

Given the ever increasing number of video card benchmarks coming out nowadays, you'd assume that you have a lot more numbers to worry about. Well, not exactly. Some of these benchmarks take a look at the frame rates recorded during runs while some look at the number of times a certain type of calculation (related to rendering a frame) can be repeated within a given time frame. At the end, a lot of benchmarks run multiple types of tests and the score obtained is unique to that benchmark only. This is why, you always need to compare two graphics cards' benchmark scores for the same benchmark which was run at exactly the same settings. This makes scores highly contextual. 